

Practice Problem Sheet-2
Transform Calculus
(MA20202)

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1. Find $L^{-1} \left\{ \frac{s}{(s+1)^5} \right\}$

Ans: $\frac{e^{-t}}{24} (4t^3 - t^4)$

2. Find $L^{-1} \frac{s+3}{s(s-1)(s+2)}$

Ans: $-\frac{3}{2} + \frac{4}{3}e^t + \frac{1}{6}e^{-2t}$

3. Find $L^{-1} \frac{s-1}{s^2+2s-8}$

Ans: $\frac{1}{6}e^{2t} + \frac{5}{6}e^{-4t}$

4. Find $L^{-1} \frac{e^{-7s}}{(s-3)^3}$

Ans: $\frac{1}{2}u(t-7)(t-7)^2 e^{3(t-7)}$

5. Find $L^{-1} \frac{s^2}{s^2+1}$

Ans: $s(t) - \sin t$

6. Find $L^{-1} \left\{ \frac{e^{4-3s}}{(s+4)^{5/2}} \right\}$

Ans: $\frac{4}{3\sqrt{\pi}} (t-3)^{3/2} e^{-4(t-3)} u(t-3)$

7. Find $L^{-1} \left\{ \frac{s^2-1}{(s^2+1)^2} \right\}$

Ans: $t \cos t$

Hence find $L^{-1} \left\{ \frac{9s^2-1}{(9s^2+1)^2} \right\}$

$\frac{t}{9} \cos\left(\frac{t}{3}\right)$

8. If $L^{-1} \left\{ \frac{s}{(s^2+1)^2} \right\} = \frac{1}{2} t \sin t$, find $L^{-1} \left\{ \frac{32s}{(16s^2+1)^2} \right\}$

Ans: $\frac{t}{4} \sin \frac{t}{4}$

9. If $L^{-1} \left\{ \frac{e^{-1/s}}{\sqrt{s}} \right\} = \frac{\cos 2\sqrt{t}}{\sqrt{\pi t}}$, find $L^{-1} \left\{ \frac{e^{-\frac{a}{s}}}{s^{1/2}} \right\}$

Ans: $\frac{\cos 2\sqrt{at}}{\sqrt{\pi t}}$

10. Prove that $L^{-1} \left\{ \frac{s}{s^4+s^2+1} \right\} = \frac{2}{\sqrt{3}} \sinh \frac{t}{2} \sin \frac{t\sqrt{3}}{2}$

11. Prove that $L^{-1} \log \frac{s+3}{s+2} = \frac{1}{t} [e^{-2t} - e^{-3t}]$

12. Prove that $L^{-1} \left\{ \frac{s}{(s^2+a^2)^2} \right\} = \frac{1}{2a} t \sin at$

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